

Electric Vehicle Cars – Comprehensive Analysis of Specifications, Battery Performance, Driving Range and Efficiency

1. Project Overview and Objective

This project focuses on cleaning, preprocessing, analysing, and visualizing electric vehicle (EV) data using Python. The analysis covers EV specifications such as battery capacity, motor power, vehicle weight, motor efficiency, certified range, and real-world driving range to identify meaningful patterns and relationships.

Objective:

The main objective is to analyse EV specifications and identify important factors associated with real-world driving range. The project also aims to understand relationships between battery capacity, motor power, vehicle weight, efficiency, and driving conditions using statistical analysis and visualizations.

2. Data Sources

- **Source Description and Timeline:** Hugging Face – India EV Range Dataset, provided by SohamThakkar-07. The original dataset contains 7,291 rows and 25 features. For this project, 800 rows and 12 original columns were selected. The dataset represents electric vehicle specifications and driving-condition data; a specific collection period is not specified.
 - **Domain:** Automotive / Electric Vehicle (EV) Industry
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3. Problem Statement

- To analyse electric vehicle specifications and identify meaningful patterns in EV performance.
- To study the relationship between battery capacity, motor power, vehicle weight, efficiency, and real-world driving range.
- To identify the factors associated with differences in real-world EV driving range.
- To compare EV characteristics and understand their impact on driving range using data analysis and visualizations.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Make	String / Categorical	Name of the EV manufacturer or brand
Model	String / Categorical	Model name of the electric vehicle
Variant	String / Categorical	Specific variant or version of the EV
Battery Capacity (kWh)	Numeric (Float)	Energy storage capacity of the EV battery
Motor Power (kW)	Numeric (Integer)	Power output of the electric motor
Vehicle Weight (kg)	Numeric (Integer)	Weight of the electric vehicle
Motor Efficiency (%)	Numeric (Integer)	Motor or drivetrain efficiency percentage
ARAI Certified Range (km)	Numeric (Integer)	Official certified driving range of the EV
Average Driving Speed (km/h)	Numeric (Float)	Average driving speed under the recorded conditions
Ambient Temperature (°C)	Numeric (Float)	Temperature during the driving condition
Traffic Condition	Numeric (Integer)	Traffic condition level in the recorded driving condition
Real World Range (km)	Numeric (Float)	Real-world driving range of the EV

5. Tools & Technologies

- **Python:** Data loading, cleaning, preprocessing, statistical analysis, and visualization.
- **Pandas:** Data manipulation, cleaning, and DataFrame operations.
- **NumPy:** Numerical calculations and feature transformation.
- **Matplotlib & Seaborn:** Data visualization and exploratory data analysis.
- **Google Colab:** Python code execution and analysis environment.

- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.
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6. Data Pre-Processing (Excel / Power Query)

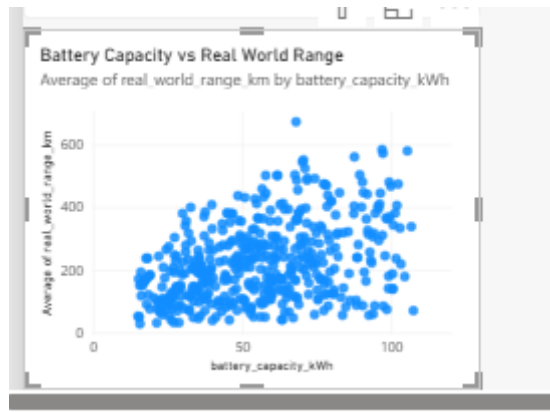
Tasks Performed:

- **Data Loading:** Loaded the EV dataset using Pandas `read_csv()`.
 - **Missing Value Check:** Checked for missing values; no missing values were found.
 - **Duplicate Check:** Checked and removed duplicate records; no duplicates were found.
 - **Data Type Checking:** Verified column names and data types.
 - **Outlier Detection:** Used the IQR method to identify unusual values in numerical columns.
 - **Skewness Analysis:** Checked skewness and identified higher positive skewness in motor power and average driving speed.
 - **Log Transformation:** Applied `log1p()` transformation to motor power and average driving speed and created transformed features.
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7.7. Visualization

- **Distribution of EV Battery Capacity – Histogram / Column Chart**
- **Distribution of EV Motor Power – Histogram / Column Chart**
- **Distribution of EV Vehicle Weight – Histogram / Column Chart**
- **Battery Capacity vs Real World Range – Scatter Plot**
- **Traffic Condition vs Real World Range – Column Chart**
- **Motor Efficiency vs Real World Range – Scatter Plot**
- **ARAI Certified Range vs Real World Range – Scatter Plot**
- **Vehicle Weight vs Real World Range – Scatter Plot**

SCREENSHOT OF VISUALIZATION



- **Battery Capacity vs Real World Range:**
 - The scatter plot shows a **positive relationship** between battery capacity and real-world driving range. In general, EVs with higher battery capacity tend to achieve higher real-world range. However, the points are widely distributed, showing that **battery capacity alone does not determine range** and other factors also influence EV performance.
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8. Dashboard (Power BI)

Dashboard 1 – EV Specifications and Range Analysis

- Total EV Records
- Average Real World Range
- Average Battery Capacity
- Average Motor Power
- EV specification distributions and Battery Capacity vs Real World Range visualization

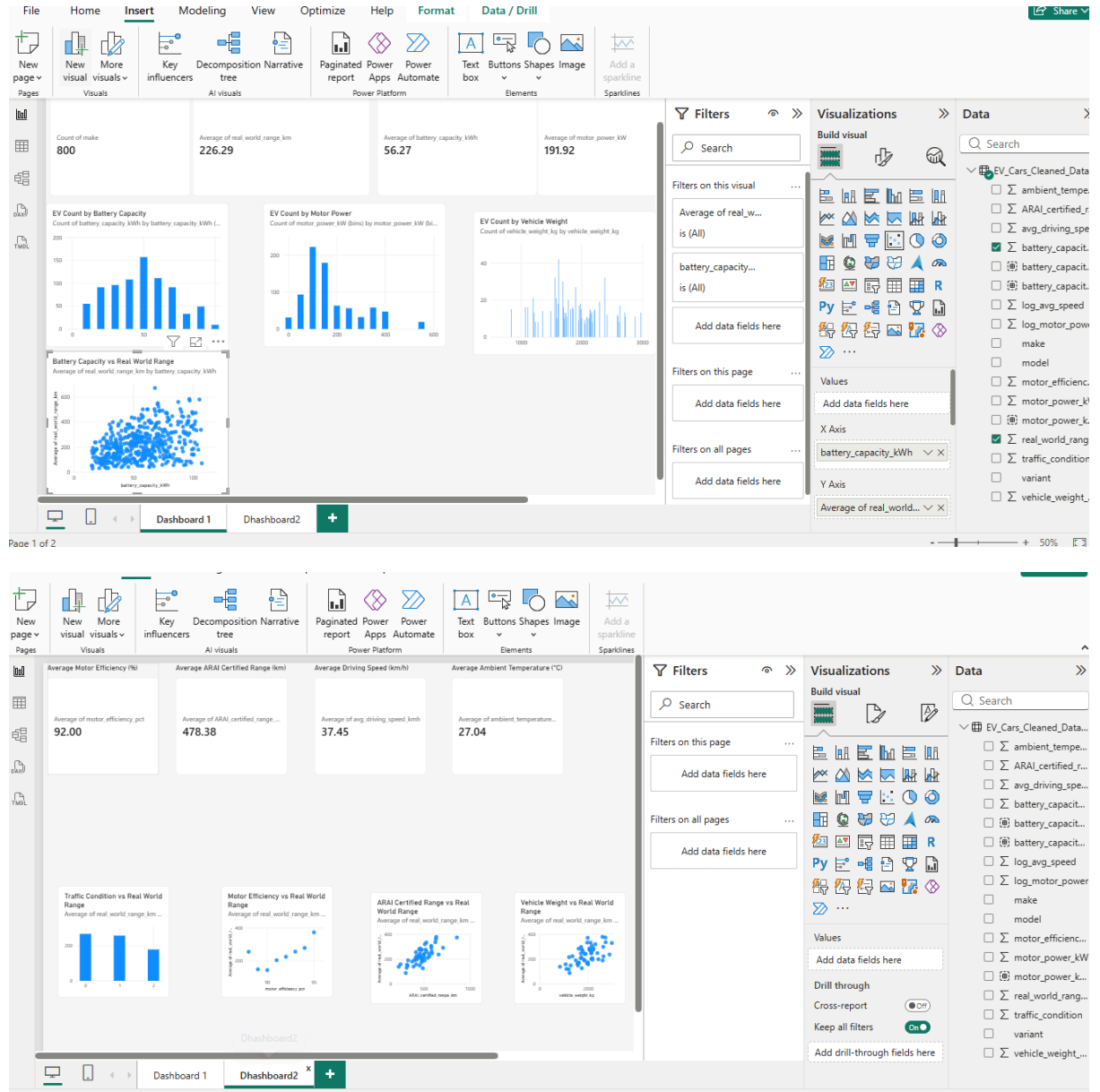
Dashboard 2 – EV Performance and Driving Conditions Analysis

- Average Motor Efficiency
- Average ARAI Certified Range
- Average Driving Speed
- Average Ambient Temperature
- Traffic, efficiency, certified range and vehicle weight vs real-world range visualizations

Interactive Features:

- Filters and slicers for interactive data exploration
- Clear titles and labels for all visuals
- Consolidated Power BI report containing both dashboards

SCREENSHOT OF DASHBOARD /REPORT



9. Insights & Conclusions

Key Insights:

- Higher battery capacity generally tends to provide higher real-world driving range.
- Battery capacity alone does not determine the real-world range.
- Motor power does not show a strong direct relationship with real-world range.

- Vehicle weight and motor efficiency influence EV performance.
- Traffic conditions and average driving speed can affect real-world range.
- ARAI certified range and real-world range may differ under practical driving conditions.

Conclusion:

The project shows that real-world EV driving range depends on multiple factors rather than a single specification. Battery capacity has a positive association with range, while vehicle weight, motor efficiency, speed, temperature, and traffic conditions also contribute to EV performance. The Python analysis and Power BI dashboards help in understanding these patterns and comparing EV performance effectively.

10. Conclusions

The integration of **Python and Power BI** proved effective for end-to-end EV data analysis, from raw data cleaning and preprocessing to interactive dashboard reporting. The project helped identify important relationships between **EV specifications, driving conditions, and real-world driving range**. The dashboards provide a clear and interactive way to understand and compare EV performance.